ITA0328 - MOBILE COMPUTING

ASSIGNMENT 3 – CO3

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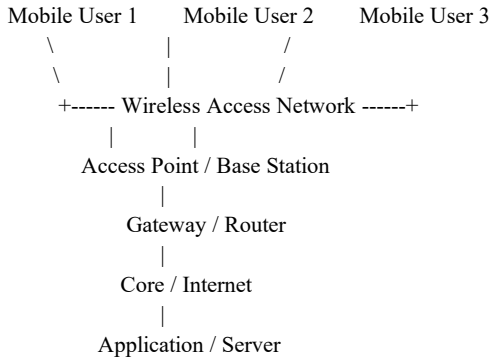
Q1 – Wireless Network Architecture and Mobility Analysis (6 Marks)

1.1 Introduction to Wireless Networks

A wireless network connects devices through radio waves instead of physical cables. It allows users to communicate while moving within a coverage area. A typical wireless network contains mobile nodes, access points or base stations, gateways, and communication links.

- Mobile nodes: smartphones, laptops, tablets, ambulances and other devices that send or receive data.
- Wireless access point/base station: provides radio access and connects wireless devices to the wider network.
- Gateway/router: forwards packets between the wireless access network and other networks.
- Communication link: radio connection between a mobile node and an access point, base station or neighbouring node.
- Network services: authentication, addressing, routing, mobility management and security.

1.2 Wireless Network Architecture



In this infrastructure-based architecture, mobile users communicate with the access point or base station. The infrastructure forwards their packets through a gateway to the destination network. The reverse path carries responses back to the mobile user.

1.3 Wireless Communication Characteristics and Constraints

Characteristic	Explanation / Effect
Limited bandwidth	Radio spectrum is shared by many users; available data rate may be lower than a wired link.
Variable signal strength	Distance, obstacles and interference can weaken the radio signal.
Connectivity changes	A user may enter or leave coverage, causing temporary disconnection.
Mobility	A moving device may need to change its access point or base station.
Interference and errors	Other transmitters and environmental noise may cause corrupted or lost packets.
Energy limitation	Battery-powered mobile devices must use radio and processing resources efficiently.

1.4 Communication and Mobility Scenario

Consider a user travelling from Cell A to Cell B. Initially, the mobile node is connected to Base Station A. As the user moves away, the signal from A decreases and the signal from B becomes stronger. The network performs a handoff so that the mobile node can continue communication through B.

Time 1: Mobile Node ---- Base Station A ---- Core Network
Time 2: Mobile Node moves toward Cell B; A signal decreases
Time 3: Mobile Node ---- Base Station B ---- Core Network

1.5 Handoff, Packet Loss and Delay

- Handoff: transfer of an active connection from one access point/base station to another.
- Handoff interruption: a short period during which packets may be delayed or lost while the new connection is established.
- Packet loss: packets may be lost because of weak signal, congestion, link failure or handoff.
- Transmission delay: time required for a packet to travel from sender to receiver; it includes transmission, propagation, processing and queuing delays.
- Increasing mobility: faster movement can cause more frequent handoffs and may increase signalling, delay and packet loss.

1.6 Illustrative Mobility Analysis

The following values are an example for explaining the analysis; they are not measured experimental results.

Condition	Handoffs in 1 minute	Average delay (ms)	Packet delivery
Low mobility	0–1	40	98%
Medium mobility	2–3	65	95%
High mobility	5–6	110	89%

The example indicates that frequent movement can increase handoff frequency and delay and reduce packet delivery. Actual results depend on speed, coverage overlap, network load and handoff implementation.

1.7 Mobility Decision Sequence (Pseudocode)

```
ALGORITHM ManageWirelessMobility(MN, CurrentAP, CandidateAP)
INPUT: Mobile node MN, current access point, candidate access point
OUTPUT: Handoff result and active point of attachment
BEGIN
  MEASURE signal quality of CurrentAP and CandidateAP
  IF CandidateAP is reachable AND
    SignalQuality(CandidateAP) > SignalQuality(CurrentAP) THEN
      AUTHENTICATE MN with CandidateAP
      ESTABLISH new wireless link
      UPDATE network attachment information
      IF new link is successful THEN
        SWITCH traffic to CandidateAP
        RELEASE old link
        RETURN HANDOFF_SUCCESS
      ELSE
        KEEP CurrentAP if it is still available
        RETURN HANDOFF_FAILURE
      END IF
  ELSE
      CONTINUE communication through CurrentAP
      RETURN NO_HANDOFF
  END IF
END
```

1.8 Conclusion

Wireless networks provide flexible communication for mobile users, but bandwidth, interference, signal variation and movement affect performance. Efficient handoff decisions, reliable links and monitoring of delay and packet delivery help maintain communication quality.

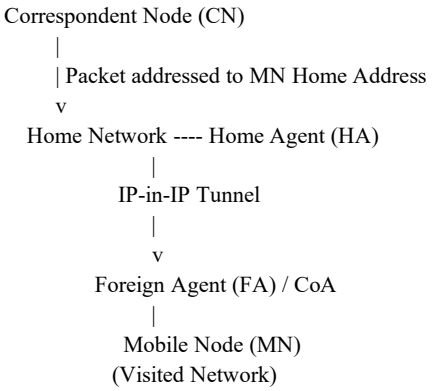
Q2 – Mobile IP and Mobility Management (7 Marks)

2.1 Mobile IP Architecture and Terminology

Mobile IP allows a mobile node to move from its home network to a visited network while retaining its permanent home IP address. The Home Agent maintains the current location binding and forwards packets to the mobile node's Care-of Address (CoA).

Entity	Role
Mobile Node (MN)	The device that changes its network attachment while maintaining communication.
Home Agent (HA)	Router on the home network that stores the MN home-address-to-CoA binding and tunnels packets.
Foreign Agent (FA)	In traditional Mobile IPv4, an agent in the visited network that assists the MN with care-of address and packet delivery.
Correspondent Node (CN)	The device or host communicating with the mobile node.
Care-of Address (CoA)	Address representing the mobile node's current point of attachment in the visited network.

2.2 Mobile IP Architecture Diagram



The CN sends packets to the MN's permanent home address. The HA intercepts them, encapsulates them and tunnels them to the registered CoA. The packet is decapsulated at the tunnel endpoint and delivered to the MN.

2.3 Agent Discovery and Registration

1. The MN receives agent advertisements or sends an agent solicitation to discover mobility agents.
2. The MN determines whether it is on its home network or a visited network.
3. In a visited network, the MN obtains a CoA, either through a Foreign Agent or another supported address mechanism.
4. The MN sends a registration request containing its home address and CoA to the HA.
5. The HA authenticates the request and creates or updates the binding.
6. The HA sends a registration reply. If accepted, packet forwarding to the CoA can begin.

2.4 Packet Delivery and Tunnelling

- The CN sends an IP packet to the MN's home address.
- The HA receives/intercepts the packet and checks its mobility binding.
- The HA encapsulates the original packet inside a new IP packet addressed to the CoA.
- The tunnel carries the encapsulated packet to the visited network.
- The tunnel endpoint decapsulates the packet and delivers the original packet to the MN.

2.5 Registration and Packet-Flow Pseudocode

```
ALGORITHM RegisterAndForward(MN, HA, CN, CurrentNetwork)
```

```
INPUT: MN, HA, CN and the MN's current network
OUTPUT: Registration status and packet-delivery status
BEGIN
  IF CurrentNetwork = MN.HomeNetwork THEN
     CoA <- MN.HomeAddress
     RegistrationStatus <- HOME
  ELSE
     CoA <- OBTAIN_CARE_OF_ADDRESS(CurrentNetwork)
     SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA
     WAIT FOR RegistrationReply
     IF Reply = ACCEPTED THEN
        HA.Binding[MN.HomeAddress] <- CoA
        RegistrationStatus <- REGISTERED
     ELSE
        RETURN REGISTRATION_FAILURE
     END IF
  END IF
  CN SENDS packet to MN.HomeAddress
  HA LOOKS UP HA.Binding[MN.HomeAddress]
  IF a valid binding exists THEN
     ENCAPSULATE packet and FORWARD it to CoA
     DECAPSULATE at tunnel endpoint
     DELIVER packet to MN
     RETURN DELIVERY_SUCCESS
  ELSE
     DROP packet
     RETURN DELIVERY_FAILURE
  END IF
END
```

2.6 Mobile IP Challenges

Challenge	Effect	Possible reduction
Registration delay	Communication may pause while the MN registers its new CoA.	Use timely agent discovery, efficient registration and suitable registration lifetimes.
Tunnelling overhead	Encapsulation adds header bytes and processing.	Use efficient tunnelling and avoid unnecessary encapsulation where supported.
Triangular routing	Packets from CN travel through HA, possibly taking a longer path.	Use route optimisation where supported and secure.
Packet loss during handoff	Packets may be lost while the old path is broken and the new path is established.	Buffer packets, use reliable transport/retransmission and minimise handoff time.
Security	False registration could redirect traffic to an attacker.	Authenticate registration messages and protect mobility signalling.
Scalability	Large numbers of mobile nodes create binding and signalling load at agents.	Use scalable agent design and control unnecessary updates.

2.7 Frequent Network Changes During an Active Session

Frequent changes can cause repeated agent discovery, registration signalling, tunnel updates and temporary packet loss. The HA may briefly hold an outdated CoA, so packets can be sent to the previous location. Delay and control overhead increase, and triangular routing may add further latency.

These problems can be reduced by using fast and reliable handoff procedures, overlapping coverage where possible, buffering packets during the transition, promptly updating the HA binding, using route optimisation when appropriate, and

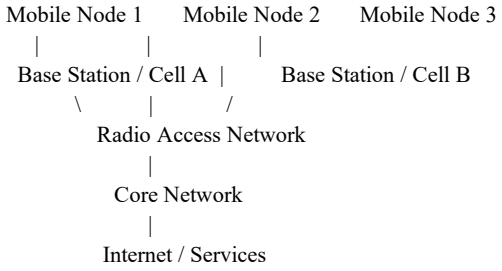
authenticating all registration messages. The network should avoid unnecessary registration updates and monitor packet loss and delay.

2.8 Conclusion

Mobile IP supports session continuity by separating the permanent home address from the current point of attachment. Agent discovery, registration, binding updates and tunnelling enable packet delivery, while efficient handoff and security mechanisms reduce mobility-related problems.

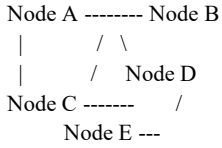
Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 Marks)

3.1 Cellular Network Architecture



A cellular network is infrastructure-based. Mobile devices communicate through base stations, which connect them to the core network. The network manages coverage, authentication, resource allocation and handoff between cells.

3.2 Ad Hoc Network Architecture



An ad hoc network is formed by wireless nodes without requiring fixed base stations or access points. Each node can act as both a host and a router. Packets may travel through multiple intermediate nodes using distributed routing.

3.3 Technical Comparison

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Requires base stations and core infrastructure.	No fixed infrastructure is required.
Communication	Normally through base stations and network infrastructure.	Direct node-to-node or multi-hop communication.
Routing	Infrastructure-assisted and centrally coordinated.	Distributed routing among mobile nodes.
Mobility	Uses cellular mobility and handoff management.	Uses route discovery and route maintenance as topology changes.
Topology	Planned cell structure; base stations are fixed.	Dynamic; links and neighbours change frequently.
Scalability	Can support large user populations with planned capacity.	Scalability may be limited by wireless contention and routing overhead.
Handoff	Transfers a device between cells/base stations.	No cellular handoff; routes and neighbours are updated.
Failure handling	Failure of a base station or core component can affect service.	Alternate peer routes may allow communication if nodes remain connected.
Applications	Public mobile networks, broadband and large-area coverage.	Disaster response, military, temporary and field networks.

3.4 Advantages and Limitations

Cellular networks provide broad planned coverage, managed resources and generally predictable service where infrastructure is available. Their main limitation in a disaster is dependence on functioning base stations, backhaul and power.

Ad hoc networks can be deployed quickly and may continue operating without fixed infrastructure. They can route around some failed links through alternate nodes. However, performance depends on node density, battery power, radio range, interference and the availability of suitable multi-hop routes.

3.5 Emergency / Disaster Network Selection

For an emergency area where fixed infrastructure is unavailable, an ad hoc wireless network is the more suitable primary choice. Rescue personnel, ambulances and police vehicles can form a temporary multi-hop network. Each device forwards packets for other devices, allowing communication without a working base station.

If some cellular infrastructure remains operational, it can be used as an additional connection to external services. A hybrid design is therefore useful: ad hoc communication locally and cellular/satellite backhaul when available.

3.6 Architecture Selection Pseudocode

```
ALGORITHM SelectNetwork(InfraAvailable, ImmediateDeployment, MultiHopNeeded)
INPUT: Infrastructure availability, deployment urgency, multi-hop requirement
OUTPUT: Selected network architecture
BEGIN
  IF InfraAvailable = TRUE THEN
     Architecture <- CELLULAR
  ELSE IF ImmediateDeployment = TRUE OR MultiHopNeeded = TRUE THEN
     Architecture <- AD_HOC
  ELSE
     Architecture <- HYBRID_OR_ALTERNATIVE
  END IF
  RETURN Architecture
END
```

3.7 Conclusion

Cellular networks are suitable when fixed infrastructure and wide-area managed coverage are available. Ad hoc networks are suitable for rapidly deployed environments without infrastructure. In disaster conditions, an ad hoc or hybrid network offers better resilience.

Q4 – Applications of Mobile IP and Routing Requirements (7 Marks)

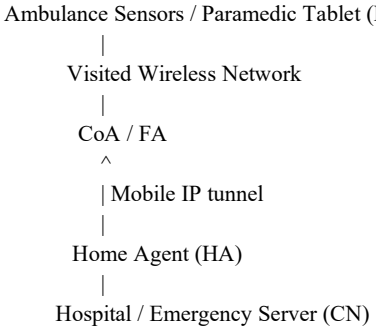
4.1 Selected Application: Emergency Ambulance Communication

An ambulance may travel across different networks while transmitting patient information, vital signs, location and estimated arrival time to a hospital. Mobile IP can help the ambulance retain a stable home address while changing its network attachment.

4.2 Mobile Entities and Endpoints

- Mobile node: ambulance communication gateway, paramedic tablet or mobile medical device.
- Home Agent: maintains the ambulance's home-address-to-CoA binding.
- Foreign Agent / visited network: supports attachment in a visited network in traditional Mobile IPv4.
- Correspondent node: hospital server, emergency coordination centre or another ambulance.
- Application server: receives patient data, location updates and emergency messages.

4.3 Application Network Architecture



When the ambulance moves into another coverage area, it obtains a new CoA and registers it with the HA. The hospital server continues addressing the ambulance using its stable home address. The HA forwards packets through a tunnel to the current CoA.

4.4 Communication Flow

7. The ambulance starts in its home network and communicates with the hospital server.
8. It enters a visited network and discovers the available mobility agent/network.
9. It obtains a CoA and registers the new location with the HA.
10. The HA updates the binding between the ambulance's home address and CoA.
11. Packets from the hospital server reach the HA and are tunnelled to the current location.
12. When the ambulance moves again, the binding is updated and forwarding continues through the new CoA.

4.5 Routing and Performance Requirements

Requirement	Importance in ambulance communication
Low delay	Critical alerts and patient information should reach the hospital quickly.
High packet delivery	Medical data and emergency messages should not be lost.
Continuous connectivity	Communication should continue as the ambulance crosses coverage areas.
Fast handoff	The interruption during network change should be minimised.
Security and privacy	Patient information and control messages must be protected.
Reliability	The system should detect link failure and use an alternate connection where possible.
Bandwidth adaptation	Video and large data transfers may need adjustment when bandwidth is limited.

4.6 Suitability of Mobile IP

Mobile IP is useful when the application needs a stable IP identity while the device changes networks. It supports location-independent communication and allows the hospital server to continue using the ambulance's home address. However, Mobile IP alone does not guarantee low latency or zero packet loss. The wireless access network, handoff procedure, routing protocol, security and backhaul quality also affect performance.

4.7 Performance Evaluation Method

The application can be evaluated by moving a simulated mobile node between two access networks and sending periodic packets to a hospital server. Record average end-to-end delay, packet delivery ratio, handoff interruption time and registration time. Compare the results before, during and after handoff.

```
ALGORITHM EvaluateMobileApplication(MN, HA, Server)
INPUT: Mobile node, Home Agent and hospital server
OUTPUT: Delivery and mobility performance measurements
BEGIN
  INITIALISE Sent, Received, TotalDelay, HandoffCount
  WHILE application is active DO
     SEND sequence-numbered packet from MN to Server
     IF packet reaches Server THEN
        Received <- Received + 1
        ADD measured delay to TotalDelay
     END IF
     IF new network attachment is detected THEN
        HandoffCount <- HandoffCount + 1
        OBTAIN new CoA
        REGISTER new CoA with HA
     END IF
  END WHILE
  PacketDeliveryRatio <- Received / Sent * 100
  AverageDelay <- TotalDelay / Received
  RETURN PacketDeliveryRatio, AverageDelay, HandoffCount
END
```

If no packets are received, average delay should be reported as unavailable rather than dividing by zero.

4.8 Conclusion

Mobile IP is appropriate for mobile healthcare and ambulance applications that require a stable network identity during movement. The design must also provide fast handoff, reliable routing, security, packet-loss monitoring and adequate bandwidth.

Q5 – Proactive Routing Protocols in Mobile Networks (6 Marks)

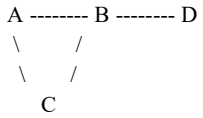
5.1 Proactive Routing Principle

A proactive routing protocol maintains routes to destinations before data transmission is requested. Nodes periodically exchange routing information and update their routing tables. Therefore, a route may already be available when an application sends a packet, reducing route-discovery delay. The disadvantage is periodic control-message overhead, even when there is little data traffic.

5.2 DSDV Operation

Destination-Sequenced Distance Vector (DSDV) is a table-driven proactive routing protocol. Each node maintains routes containing destination, next hop, hop count and destination sequence number. Sequence numbers help identify fresh route information and reduce routing loops. A route with a newer sequence number is preferred; when sequence numbers are equal, the route with the lower hop count is preferred.

5.3 Sample Topology



Assume node A reaches destination D through B. If link B–D fails, nodes exchange updated routing information. A may select an alternate route through C if a valid route is advertised.

5.4 Example Routing Table

Node	Destination	Next hop	Hop count	Sequence number	State
A	D	B	2	D: 10	VALID
A	C	C	1	C: 6	VALID
B	D	D	1	D: 10	VALID
C	D	B	2	D: 10	VALID

Illustrative topology-change example: if link B–D fails, B advertises an unreachable route for D with an updated sequence number. A then removes or invalidates the route through B. If C advertises a newer valid route to D, A can update its next hop to C.

5.5 Route Update Logic (Pseudocode)

```
ALGORITHM UpdateDSDV(Node, ReceivedRoute)
INPUT: Node and {Destination, NextHop, HopCount, SequenceNumber}
OUTPUT: Updated routing table
BEGIN
  Destination <- ReceivedRoute.Destination
  NewHopCount <- ReceivedRoute.HopCount + 1
  NewSequence <- ReceivedRoute.SequenceNumber
  IF Destination is not in RouteTable THEN
     ADD route using received next hop, hop count and sequence number
  ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
     REPLACE existing route
  ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
        NewHopCount < RouteTable[Destination].HopCount THEN
     REPLACE existing route
  ELSE
     KEEP existing route
  END IF
```

```
    RETURN RouteTable
END
```

5.6 Performance Investigation

Parameter	Expected behaviour / interpretation
Routing-control overhead	Periodic updates consume bandwidth and increase as the network or update information grows.
Route availability	Routes are usually available before a data packet is generated.
Convergence	After a topology change, nodes need time to receive and process updates.
Control messages	Periodic and triggered updates maintain current routing information.
Network size	Larger networks can require more routing information and control traffic.
Delay	Initial route-discovery delay is low, but stale routes during convergence may cause delay or loss.
Packet delivery	Good when routing information is current; mobility can temporarily invalidate routes.

5.7 Conclusion

DSDV is useful when low route-discovery delay is important and topology changes are not excessively frequent. Its sequence-number mechanism supports fresh route selection, but periodic updates create overhead, especially in large or highly mobile networks.

Q6 – Reactive Routing and Protocol Comparison (7 Marks)

6.1 Reactive Routing Principle

Reactive routing is demand-driven. A route is discovered only when a source needs to send data and does not already have a valid route. This reduces periodic control traffic but introduces an initial route-discovery delay.

6.2 AODV Operation

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. It uses Route Request (RREQ), Route Reply (RREP) and Route Error (RERR) messages.

- RREQ: broadcast by the source to discover a route to a destination.
- RREP: sent by the destination, or an intermediate node with a sufficiently fresh route, to establish the route back to the source.
- RERR: informs affected nodes that a route or link is no longer available.

6.3 Route Discovery and Maintenance

13. The source checks its routing table for a valid route.
14. If no valid route exists, it creates and broadcasts an RREQ with a request identifier.
15. Intermediate nodes record a reverse route toward the source and forward the RREQ if they have not already processed it.
16. The destination or an intermediate node with a valid fresh route sends an RREP.
17. The RREP travels back using the reverse path, establishing forward-route information.
18. During communication, nodes monitor the active route. If a link fails, the node invalidates the route and sends an RERR.
19. The source may use an alternate valid route or start a new route discovery.

6.4 AODV Route Discovery Pseudocode

```
ALGORITHM AODVRouteDiscovery(Source, Destination)
INPUT: Source and destination nodes
OUTPUT: Valid route or ROUTE_NOT_FOUND
BEGIN
  IF RouteTable[Destination].State = VALID THEN
     RETURN RouteTable[Destination]
  END IF
  Generate unique RouteRequestID
  CREATE RREQ(Source, Destination, RouteRequestID)
  BROADCAST RREQ to neighbours
  WAIT for RREP or route-discovery timeout
  IF RREP received THEN
     ESTABLISH reverse path toward Source
     ESTABLISH forward path toward Destination
     MARK route as VALID
     RETURN RouteTable[Destination]
  ELSE
     RETURN ROUTE_NOT_FOUND
  END IF
END
```

6.5 Route Maintenance Pseudocode

```
ALGORITHM MaintainAODVRoute(Source, Destination)
INPUT: Active source-destination route
OUTPUT: Route status
BEGIN
  MONITOR every link on the active route
```

```
  IF LinkFailureDetected THEN
     MARK failed route as INVALID
     IF an alternate valid route exists THEN
        SELECT alternate route
        RETURN ALTERNATE_ROUTE_USED
     ELSE
        CREATE and SEND RERR to affected nodes
        IF Source still needs communication THEN
           START route discovery
        ELSE
           RETURN ROUTE_TERMINATED
        END IF
     END IF
  END IF
  RETURN ROUTE_ACTIVE
END
```

6.6 AODV versus DSDV

Feature	DSDV (Proactive)	AODV (Reactive)
Route discovery	Routes maintained continuously.	Route discovered when required.
Routing table	Maintains routes to destinations in advance.	Maintains active/discovered routes and related state.
Control overhead	Periodic updates create continuous overhead.	Control messages mainly occur during discovery and repair.
Initial communication delay	Usually low because route is already known.	Can be higher while RREQ/RREP discovery completes.
Mobility handling	Routing tables must be updated as topology changes.	Broken active routes trigger repair or rediscovery.
Infrequent traffic	May waste bandwidth maintaining unused routes.	Often more efficient because routes are found on demand.
Typical use	Networks where traffic is frequent and low route setup delay is important.	Networks with intermittent traffic and changing topology.

6.7 Protocol Selection for a Highly Dynamic Network

For a highly dynamic mobile network with frequent topology changes, AODV is generally the more suitable choice when traffic is intermittent or bandwidth is limited. It avoids continuously advertising routes that may quickly become stale and discovers routes when needed. It also uses route-error messages and route rediscovery to respond to link failures.

However, AODV introduces route-discovery delay when a source has no valid route. If the application generates continuous high-rate traffic and requires very low route setup delay, a proactive approach such as DSDV may be preferable, provided the network can tolerate its periodic overhead. The final choice depends on traffic rate, mobility, bandwidth and delay requirements.

6.8 Conclusion

AODV reduces unnecessary periodic updates and is suitable for many dynamic, intermittently used mobile networks. DSDV offers routes in advance and lower initial route setup delay, but consumes bandwidth to maintain routing tables.

Q7 – Design a Mobile Emergency Communication Network (10 Marks)

7.1 Problem Statement and Design Objectives

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously. Fixed communication infrastructure may be damaged or unavailable. The proposed network must provide rapid deployment, reliable local communication, low delay for critical messages, support for mobility and recovery from node or link failures.

7.2 Stage 1 – Independent Design Proposals

Student A – Architecture and Mobility Proposal

- Use a wireless ad hoc network among rescue vehicles, portable devices and field teams because it can operate without fixed base stations.
- Use multi-hop communication so a device can forward packets for another device that is outside direct radio range.
- Use Mobile IP when a mobile node must retain a stable home address while moving between available IP networks.
- Where cellular or satellite connectivity is available, use it as an external gateway to hospitals and emergency coordination centres.
- Use handoff and location registration to update the mobile node's current point of attachment.

Student B – Routing and Failure Proposal

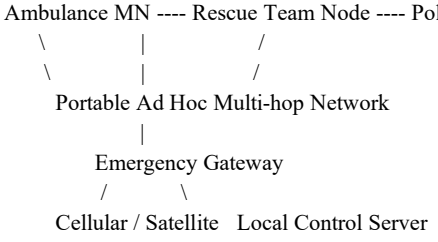
- Select AODV as the primary routing protocol for a changing, intermittently used emergency ad hoc network.
- Use RREQ/RREP for on-demand route discovery and RERR for reporting broken routes.
- Monitor active links and maintain alternate routes when available.
- If a node or link fails, invalidate the route and initiate repair or rediscovery.
- Evaluate end-to-end delay, packet delivery ratio, routing overhead and route recovery time.

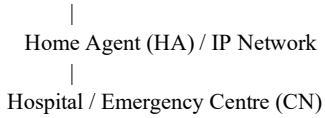
7.3 Stage 2 – Comparison of the Two Designs

Aspect	Student A proposal	Student B proposal
Main focus	Network architecture and mobility continuity.	Routing, route maintenance and failure recovery.
Strength	Supports infrastructure-less local communication and stable addressing where Mobile IP is deployed.	AODV discovers routes on demand and reacts to link failures.
Limitation	Mobile IP needs reachable agents/registration infrastructure for its conventional operation; it does not itself create ad hoc routes.	Route discovery can add delay; frequent failures can cause repeated control messages.
Combined contribution	Defines the nodes, access networks and mobility mechanism.	Provides packet paths between nodes and recovery when paths break.

7.4 Stage 3 – Joint Final Network Design

The final design uses a hybrid emergency network. An ad hoc multi-hop network provides local communication among ambulances, police, rescue teams and medical personnel. A surviving cellular or satellite gateway connects the field network to hospitals and the emergency control centre. If fixed infrastructure is completely unavailable, local ad hoc communication continues and messages can be forwarded when a gateway becomes available.





The ad hoc routing protocol moves packets across the local network. Mobile IP is used at the IP mobility boundary when a mobile node moves between supported IP networks and needs to retain a stable home address. The two functions are complementary: Mobile IP manages a node's network location, while AODV finds paths through the ad hoc topology.

7.5 Mobile IP Components and Registration

- Mobile Node: ambulance gateway or command vehicle that moves between networks.
- Home Agent: maintains the binding between the MN's home address and current CoA.
- Foreign Agent / visited network: supports the MN's attachment in traditional Mobile IPv4.
- Correspondent Node: hospital or emergency control centre.
- Registration: after obtaining a CoA, the MN registers it with the HA; the HA updates the binding.

7.6 Registration and Tunnelling Process

20. The mobile node detects that it has entered a visited network.
21. It discovers the available mobility agent/network and obtains a CoA.
22. It sends a registration request to the HA.
23. The HA authenticates the request and updates the home-address-to-CoA binding.
24. A hospital/control-centre packet addressed to the home address reaches the HA.
25. The HA encapsulates and tunnels the packet to the CoA, where it is decapsulated and delivered.

7.7 Selected Routing Protocol: AODV

AODV is selected for the local ad hoc network because the disaster environment has changing node positions and intermittent traffic. It does not require every node to maintain periodic routes to every destination. When a rescue team needs to contact an ambulance, it initiates route discovery. If a link breaks, an RERR informs affected nodes and a new route can be discovered.

AODV may introduce initial discovery delay and repeated control overhead when links fail frequently. For continuous high-rate traffic with strict low-delay requirements, a proactive protocol could be evaluated as an alternative. The selection should be confirmed through simulation under realistic mobility and traffic loads.

7.8 Failure Handling

Failure	Proposed response
Mobile node failure	Detect missing neighbour/hello messages; invalidate routes using the failed node and rediscover.
Wireless link failure	Generate RERR, mark the route invalid and use an alternate route if one exists.
Gateway failure	Select another gateway if available; otherwise continue local ad hoc communication and queue non-urgent external messages.
Network disconnection	Store critical messages temporarily, retry transmission and forward them when connectivity returns.
HA/registration failure	Retry registration securely; retain local communication where possible and report that external Mobile IP reachability is unavailable.
Congestion / limited bandwidth	Prioritise emergency voice, alerts and essential patient data; defer or reduce non-critical traffic.

7.9 Stage 4 – Validation and Performance Evaluation

A network simulator such as NS-3 or an equivalent mobile ad hoc network simulator can be used. Create a field area with ambulances, police vehicles, rescue teams and a gateway. Vary the number of nodes and mobility speed. Generate emergency messages and record the following metrics.

Metric	Measurement
End-to-end delay	Average time from packet generation at source to reception at destination.
Packet delivery ratio (PDR)	(Packets received / Packets sent) × 100%.
Routing overhead	Number of routing-control packets or control bytes transmitted.
Handoff / registration delay	Time from detecting a new network to successful registration and usable forwarding.
Route recovery time	Time between link failure detection and restoration of a usable route.
Connectivity availability	Percentage of the test duration during which a usable path exists.

7.10 Test Cases and Illustrative Results

The table below is a sample reporting format with illustrative values only. It is not a claim that a simulator or physical experiment has been run.

Test case	Nodes	Mobility	Average delay (ms)	PDR (%)	Routing overhead (control packets)
T1	10	Low	45	98	120
T2	20	Medium	70	94	260
T3	30	High	115	87	510

Interpretation: as node count and mobility increase, route changes and control signalling may increase. Delay may rise and packet delivery may fall because links break more often and routes require repair. Actual values must be replaced with measured simulator results before presenting this table as experimental evidence.

7.11 Stage 5 – Cross-Defence

Why is AODV appropriate? AODV discovers routes only when needed and reacts to broken links through route errors and rediscovery. This reduces the need for continuous network-wide route updates in an intermittently used, dynamic emergency network.

How does Mobile IP maintain communication? Mobile IP keeps the permanent home address associated with the mobile node. When the node moves, it obtains a new CoA and registers it with the HA. The HA tunnels packets addressed to the home address to the current CoA.

7.12 Stage 6 – Joint Evaluation and Design Limitations

- The ad hoc architecture can be deployed rapidly and does not depend on fixed local base stations.
- Mobile IP is useful when communication must continue across different IP networks using a stable home address; it is not mandatory for purely local ad hoc communication.
- AODV supports dynamic route discovery and repair, but discovery and repeated failures can increase delay and overhead.
- Packet delivery depends on node density, radio range, interference, available energy and route stability.
- When no end-to-end path exists, packets cannot be delivered immediately; store-and-forward or message queuing can help when connectivity returns.

- Security is essential: authenticate users and routing/mobility messages, encrypt sensitive patient information and control access to emergency services.
- The design may be unsuitable if node density is too low, all gateways fail for a long period, devices have insufficient battery power, or the application requires guaranteed real-time delivery beyond the network's capability.

7.13 Final Conclusion

Before selecting Mobile IP and a routing protocol for a highly mobile wireless environment, a designer should analyse infrastructure availability, node density, mobility speed, topology-change rate, traffic pattern, bandwidth, delay limits, packet-loss tolerance, energy, security, scalability and failure-recovery needs. Mobile IP should be selected when stable IP identity across networks is required. AODV is suitable for many dynamic networks with intermittent traffic, while DSDV may be preferred when routes must be ready before transmission and periodic overhead is acceptable. Simulation and measured performance are necessary to validate the final design.

OVERALL CONCLUSION

This assignment explains wireless network architecture, mobility management, Mobile IP, cellular and ad hoc networks, mobile applications, and proactive and reactive routing protocols. Wireless communication is affected by bandwidth limits, interference, node movement and link failures. Mobile IP supports location-independent communication through registration and tunnelling. Cellular networks provide managed wide-area coverage, while ad hoc networks support rapid infrastructure-less deployment. DSDV maintains routes proactively, whereas AODV discovers routes on demand. For a disaster-response environment, a hybrid architecture using local ad hoc communication, AODV routing and Mobile IP at supported network boundaries can provide a flexible solution. The design must be validated using delay, packet delivery ratio, routing overhead and recovery measurements.

REFERENCES / BASIS

1. ITA03-MP Assignment 3 – Mobile Computing, CO3, supplied assignment document.

2. Mobile Computing pseudocode and algorithmic standard included in the supplied assignment document.

Note: Architecture diagrams and sample numerical tables in this answer are explanatory representations. Numerical performance values are illustrative and should be replaced with actual simulation or experimental measurements if required by the faculty.